

Assignment 2 Solutions

Modern Cryptography (2)

Problem 1. For the Schnorr signature scheme, reusing the same random nonce r to sign two distinct messages is *not secure*. Let the two signatures be

$$\sigma_1 = (R, z_1), \quad \sigma_2 = (R, z_2),$$

where the same commitment $R = g^r$ is used in both signatures. For Schnorr signatures,

$$z_i = r + e_i s \pmod{p},$$

where s is the signing key and $e_i = H(R, M_i)$ is the challenge for message M_i . Hence

$$z_1 - z_2 = (e_1 - e_2)s \pmod{p}.$$

If $e_1 \neq e_2$, then the secret key can be recovered as

$$s = (z_1 - z_2)(e_1 - e_2)^{-1} \pmod{p}.$$

Since $M_1 \neq M_2$, we expect $e_1 \neq e_2$ except with negligible probability (assuming the hash function behaves like a random oracle or is collision resistant in the relevant sense). Therefore, reusing the same nonce immediately reveals the secret key, and the scheme becomes completely insecure.

Problem 2. Consider the variant of DSA in which the hash function H is omitted, so the response is computed as

$$z = r^{-1}(M + ds) \pmod{p},$$

where $d = F(R)$ and $R = g^r$. We show that this scheme is existentially forgeable.

An adversary can forge a valid signature on a message of its choice as follows. Choose arbitrary values

$$a, b \in \mathbb{Z}_p, \quad b \neq 0,$$

and define

$$R := g^a h^b.$$

Then query the random oracle F at R and obtain

$$d := F(R).$$

Now set

$$z := db^{-1} \pmod{p}, \quad M := az \pmod{p}.$$

We claim that (d, z) is a valid signature on M . Indeed,

$$(g^M h^d)^{z^{-1}} = (g^{az} h^d)^{z^{-1}} = g^a h^{dz^{-1}}.$$

Because $z = db^{-1}$, we have $dz^{-1} = b$, and therefore

$$(g^M h^d)^{z^{-1}} = g^a h^b = R.$$

Also, by construction, $d = F(R)$. Hence the verification equation is satisfied. So the adversary outputs the forged signature

$$\sigma = (d, z)$$

on message M , without knowing the secret key s .

Therefore, omitting H makes the scheme insecure.

Problem 3. We show that if the Discrete Logarithm problem is hard in G , then the DSA identification scheme is UI-PA secure.

Assume toward contradiction that there exists a PPT adversary $\mathcal{A}$ that breaks UI-PA security with non-negligible advantage

$$\text{Adv}_{\text{UI-PA}}^{\mathcal{A}}(\lambda) = \Pr[\mathcal{A} \text{ successfully impersonates}] = \varepsilon(\lambda),$$

where $\varepsilon(\lambda)$ is non-negligible. We construct a PPT algorithm $\mathcal{B}$ that solves the discrete logarithm problem.

Algorithm $\mathcal{B}$ is given (G, p, g, h) with $h = g^s$, and its goal is to recover s . It runs $\mathcal{A}$ and simulates all passive transcripts requested by $\mathcal{A}$. An accepting transcript of the identification protocol has the form

$$(R, e, d, z)$$

with

$$R = (g^e h^d)^{z^{-1}}.$$

Hence $\mathcal{B}$ can generate a perfectly distributed accepting transcript by choosing

$$e, d \xleftarrow{\$} \mathbb{Z}_p, \quad z \xleftarrow{\$} \mathbb{Z}_p^*, \quad R := (g^e h^d)^{z^{-1}}.$$

This has exactly the same distribution as a real protocol transcript, so the simulation in the passive phase is perfect.

After seeing polynomially many such transcripts, $\mathcal{A}$ outputs a first message R^* . Now $\mathcal{B}$ sends a random challenge $(e, d_1) \xleftarrow{\$} \mathbb{Z}_p \times \mathbb{Z}_p$. If $\mathcal{A}$ answers with z_1 such that

$$R^* = (g^e h^{d_1})^{z_1^{-1}},$$

then $\mathcal{B}$ rewinds $\mathcal{A}$ to the point immediately after R^* was sent, keeps all internal randomness of $\mathcal{A}$ fixed, and sends a fresh independent challenge $(e, d_2) \xleftarrow{\$} \mathbb{Z}_p \times \mathbb{Z}_p$ with the same e and a new d_2 . If $\mathcal{A}$ also answers with z_2 such that

$$R^* = (g^e h^{d_2})^{z_2^{-1}},$$

then $\mathcal{B}$ extracts s .

Indeed, write $R^* = g^r$. From the two accepting transcripts we obtain

$$r \equiv (e + d_1 s) z_1^{-1} \equiv (e + d_2 s) z_2^{-1} \pmod{p}.$$

Multiplying through gives

$$(e + d_1 s) z_2 = (e + d_2 s) z_1,$$

and therefore

$$e(z_2 - z_1) = (d_2 z_1 - d_1 z_2) s \pmod{p}.$$

So whenever $d_2 z_1 - d_1 z_2 \neq 0$, we can compute

$$s = e(z_2 - z_1)(d_2 z_1 - d_1 z_2)^{-1} \pmod{p}.$$

If $d_2 z_1 - d_1 z_2 = 0$, then the above equality implies $e(z_2 - z_1) = 0$. For $e \neq 0$, this forces $z_1 = z_2$, and then from

$$e + d_1 s \equiv e + d_2 s \pmod{p}$$

we get $(d_1 - d_2)s \equiv 0$. Since $s \neq 0$ except with probability $1/p$, this would imply $d_1 = d_2$. Thus, except with probability at most $2/p$, the denominator is nonzero and extraction succeeds.

It remains to analyze the success probability. Let ω denote all randomness used by $\mathcal{A}$ and by $\mathcal{B}$ before the final challenge is chosen. For fixed ω , define

$$V(\omega, e, d) = 1$$

if and only if, when the final challenge is (e, d) , the adversary outputs a valid response z for the already fixed first message R^* . Then

$$\varepsilon = \Pr_{\omega, e, d}[V(\omega, e, d) = 1].$$

Now consider the rewinding experiment in which we sample the same ω and the same e , but two independent values $d_1, d_2 \xleftarrow{\$} \mathbb{Z}_p$. Let $\varepsilon_{\mathcal{B}}$ be the probability that both rewound runs accept and $d_1 \neq d_2$. Then

$$\varepsilon_{\mathcal{B}} \geq \Pr_{\omega, e, d_1, d_2}[V(\omega, e, d_1) = 1 \wedge V(\omega, e, d_2) = 1] - \Pr[d_1 = d_2].$$

Since $\Pr[d_1 = d_2] = 1/p$, we get

$$\varepsilon_{\mathcal{B}} \geq \sum_{\omega} \Pr[\omega] \left(\frac{1}{p} \sum_{e \in \mathbb{Z}_p} \Pr_d[V(\omega, e, d) = 1] \right)^2 - \frac{1}{p}$$

by independence of d_1, d_2 . Using Jensen's inequality (equivalently Cauchy-Schwarz),

$$\sum_{\omega} \Pr[\omega] \left(\frac{1}{p} \sum_e \Pr_d[V(\omega, e, d) = 1] \right)^2 \geq \left(\sum_{\omega} \Pr[\omega] \cdot \frac{1}{p} \sum_e \Pr_d[V(\omega, e, d) = 1] \right)^2 = \varepsilon^2.$$

Hence

$$\varepsilon_{\mathcal{B}} \geq \varepsilon^2 - \frac{1}{p}.$$

Since p is exponential in the security parameter, $1/p$ is negligible; therefore if ε is non-negligible, then $\varepsilon_{\mathcal{B}}$ is also non-negligible. Combining this with the previous extraction step, $\mathcal{B}$ solves the discrete logarithm problem with non-negligible probability, contradicting the DL assumption.

Therefore, the DSA identification scheme is UI-PA secure.

Problem 4. We prove that if the identification scheme is UI-PA secure and H, F are modeled as random oracles, then the Fiat-Shamir signature scheme is EUF-CMA secure.

The identification protocol and the Fiat-Shamir signature scheme. The public key is (G, p, g, h) , where $h = g^s$ for a secret key $s \in \mathbb{Z}_p$.

In the identification protocol, the prover first chooses $r \xleftarrow{\$} \mathbb{Z}_p$ and sends

$$R := g^r$$

to the verifier. The verifier then chooses a random challenge pair

$$(e, d) \xleftarrow{\$} \mathbb{Z}_p^2$$

and sends (e, d) to the prover. The prover replies with

$$z := r \cdot (es + d)^{-1}.$$

The verifier accepts if and only if

$$R = (g^e h^d)^{z^{-1}}.$$

The Fiat–Shamir signature scheme is obtained by replacing the verifier’s challenge by two random oracles. To sign a message M , the signer computes

$$R := g^r, \quad e := H(M), \quad d := F(R),$$

and outputs

$$z := r \cdot (es + d)^{-1}.$$

Hence the signature is (d, z) . Verification recomputes

$$e := H(M), \quad R' := (g^e h^d)^{z^{-1}},$$

and accepts if and only if

$$F(R') = d.$$

The UI-PA game. We now describe the UI-PA game more explicitly.

In the learning phase, adversary B may obtain honestly generated accepting transcripts from its UI-PA challenger. Each transcript has the form

$$(R_i, e_i, d_i, z_i)$$

such that

$$R_i = (g^{e_i} h^{d_i})^{z_i^{-1}}.$$

In the final impersonation phase, B must first send a value R^* to the UI-PA challenger. The challenger then returns a random pair

$$(e^*, d^*) \xleftarrow{\$} \mathbb{Z}_p^2,$$

and finally B must answer with z^* . Adversary B wins if

$$R^* = (g^{e^*} h^{d^*})^{(z^*)^{-1}}.$$

Reduction. Assume that there exists a PPT adversary A that makes at most q_H queries to H , at most q_F queries to F , at most q_S signing queries, and outputs a valid forgery with probability $\varepsilon(\lambda)$. We construct a PPT adversary B that breaks the UI-PA security of the identification scheme.

Adversary B runs A on input the public key (G, p, g, h) . It simulates the random oracles H and F , and it answers signing queries using the honest transcripts obtained from its UI-PA challenger.

As before, B guesses two indices uniformly at random:

$$i_H \in \{1, \dots, q_H\}, \quad i_F \in \{1, \dots, q_F\}.$$

Intuitively, B guesses that the final forgery of A will involve the i_H -th fresh query to H and the i_F -th fresh query to F .

Simulation of the random oracles. Adversary B uses lazy sampling and maintains two tables

$$T_H \quad \text{and} \quad T_F.$$

Simulation of H . Whenever A queries $H(M)$:

- if $M \in T_H$, return the recorded value;
- otherwise, if this is not the i_H -th fresh query, choose

$$e \xleftarrow{\$} \mathbb{Z}_p,$$

store $T_H[M] = e$, and return e ;

- otherwise, this is the distinguished H -query. Its value will later be set to the challenge value e^* obtained from the UI-PA challenger during the distinguished F -query.

Simulation of F . Whenever A queries $F(R)$:

- if $R \in T_F$, return the recorded value;
- otherwise, if this is not the i_F -th fresh query, choose

$$d \xleftarrow{\$} \mathbb{Z}_p,$$

store $T_F[R] = d$, and return d ;

- otherwise, this is the distinguished F -query. At this point, B sends this value R to its UI-PA challenger as the first message of the final impersonation attempt. The challenger responds with

$$(e^*, d^*) \xleftarrow{\$} \mathbb{Z}_p^2.$$

B stores

$$R^* := R, \quad T_F[R^*] = d^*,$$

and sets the answer to the distinguished H -query to be

$$e^*.$$

Thus the two special random-oracle values are programmed consistently with the same challenge pair:

$$H(M^*) = e^*, \quad F(R^*) = d^*.$$

Simulation of signing queries. Whenever A asks for a signature on a message M , adversary B obtains one fresh honest transcript

$$(R_i, e_i, d_i, z_i)$$

from its UI-PA challenger. Since this transcript is honestly generated, it satisfies

$$R_i = (g^{e_i} h^{d_i})^{z_i^{-1}}.$$

Adversary B then answers the signing query as follows:

1. If M is already in T_H with value different from e_i , then B aborts.
2. Otherwise, set

$$T_H[M] := e_i.$$

3. If R_i is already in T_F with value different from d_i , then B aborts.
4. Otherwise, set

$$T_F[R_i] := d_i.$$

5. Return the signature

$$(d_i, z_i).$$

This is a correct simulation, because for this signature the verifier computes

$$e = H(M) = e_i,$$

reconstructs

$$R' = (g^e h^{d_i})^{z_i^{-1}} = (g^{e_i} h^{d_i})^{z_i^{-1}} = R_i,$$

and then checks

$$F(R') = F(R_i) = d_i.$$

Hence the signature (d_i, z_i) is valid.

Using the forgery. Eventually A outputs a forgery $(M^\dagger, d^\dagger, z^\dagger)$ on a message $M^\dagger$ that was not previously submitted to the signing oracle. Let

$$e^\dagger := H(M^\dagger)$$

and define

$$R^\dagger := (g^{e^\dagger} h^{d^\dagger})^{(z^\dagger)^{-1}}.$$

Since the forgery verifies, we must have

$$F(R^\dagger) = d^\dagger.$$

Now consider the following good event:

1. $M^\dagger$ is exactly the distinguished i_H -th fresh H -query;
2. $R^\dagger$ is exactly the distinguished i_F -th fresh F -query;
3. no abort occurs during the simulation.

Under this event, by construction we have

$$e^\dagger = e^*, \quad d^\dagger = d^*, \quad R^\dagger = R^*.$$

Therefore,

$$R^* = (g^{e^*} h^{d^*})^{(z^\dagger)^{-1}},$$

so B can send

$$z^* := z^\dagger$$

to its UI-PA challenger and wins the impersonation game.

Probability analysis. We now estimate the success probability of B .

First, except with negligible probability, any successful forgery must involve oracle queries at both relevant points:

- A must query $H(M^\dagger)$, since otherwise $H(M^\dagger)$ is an independent random value unknown to A ;
- A must query $F(R^\dagger)$, since otherwise the condition $F(R^\dagger) = d^\dagger$ can hold only with probability at most $1/p$, which is negligible.

Conditioned on a successful forgery and on the fact that these two relevant oracle queries are made, the probability that B guesses both the correct H -query and the correct F -query is exactly

$$\frac{1}{q_H} \cdot \frac{1}{q_F} = \frac{1}{q_H q_F}.$$

Thus,

$$\Pr[B \text{ wins UI-PA}] \geq \frac{\varepsilon(\lambda)}{q_H q_F} - \text{negl}(\lambda).$$

Since q_H and q_F are polynomially bounded and $\varepsilon(\lambda)$ is non-negligible, this lower bound is also non-negligible.

This contradicts the UI-PA security of the identification scheme.